

Отчёт по лабораторной работе № 6

Название работы: Интерактивный rebase, cherry-pick, теги, reflog

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Дата выполнения: 16.05.2026

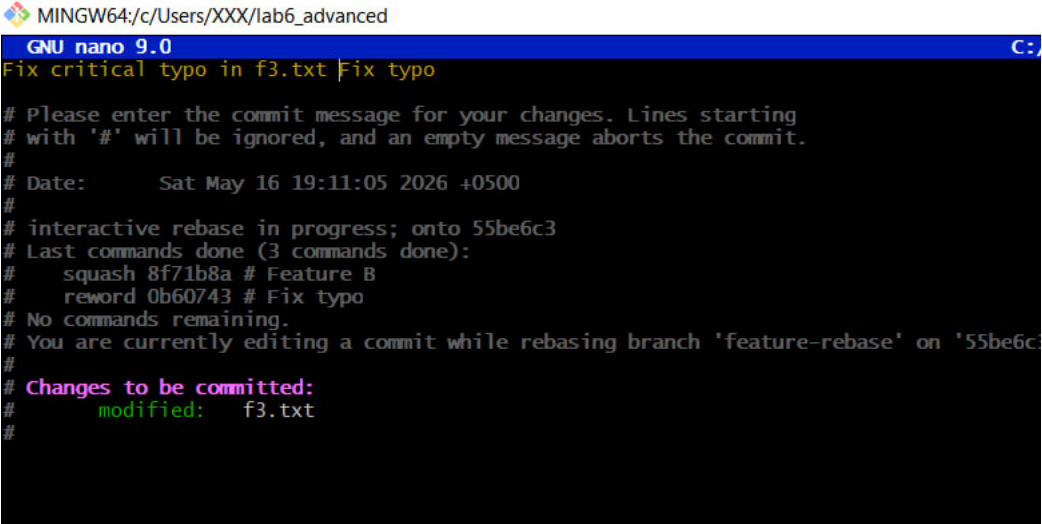
1. Выполнение шагов лабораторной работы

Отметьте галочками выполненные шаги. В столбце «Команды / вывод» напишите выполненные команды и полученный вывод (поле автоматически растягивается по содержимому).

Шаг	Описание	Вып.	Команды / вывод
1	Настроить редактор Git на nano (git config --global core.editor 'nano')	☒	XXX@DESKTOP-9CORRHG MINGW64 ~/lab6_advanced (main) \$ git config --global core.editor "nano"
2	Создать репозиторий и сделать 5 коммитов (Step 1..5)	☒	XXX@DESKTOP-9CORRHG MINGW64 / \$ cd /c/Users/XXX/lab6_advanced XXX@DESKTOP-9CORRHG MINGW64 ~/lab6_advanced (main) \$ git init Initialized empty Git repository in C:/Users/XXX/lab6_advanced/.git/ XXX@DESKTOP-9CORRHG MINGW64 ~/lab6_advanced (master) \$ git branch -M main XXX@DESKTOP-9CORRHG MINGW64 ~/lab6_advanced (main) \$ for i in {1..5}; do echo "Commit \$i" > f\$i.txt; git add .; git commit -m "Step \$i"; done warning: in the working copy of 'f1.txt', LF will be replaced by CRLF the next time Git touches it [main (root-commit) f742d99] Step 1 1 file changed, 1 insertion(+) create mode 100644 f1.txt warning: in the working copy of 'f1.txt', LF will be replaced by CRLF the next time Git touches it warning: in the working copy of 'f2.txt', LF will be replaced by CRLF the next time Git touches it [main 20f84cc] Step 2 1 file changed, 1 insertion(+) create mode 100644 f2.txt warning: in the working copy of 'f1.txt', LF will be replaced by CRLF the next time Git touches it warning: in the working copy of 'f2.txt', LF will be replaced by CRLF the next time Git touches it warning: in the working copy of 'f3.txt', LF will be replaced by CRLF the next time Git touches it [main 1389a3c] Step 3 1 file changed, 1 insertion(+) create mode 100644 f3.txt warning: in the working copy of 'f4.txt', LF will be replaced by CRLF the next time Git touches it [main 9ba0339] Step 4 1 file changed, 1 insertion(+) create mode 100644 f4.txt warning: in the working copy of 'f4.txt', LF will be replaced by CRLF the next time Git touches it warning: in the working copy of 'f5.txt', LF will be replaced by CRLF the next time Git touches it [main 31ae8ad] Step 5 1 file changed, 1 insertion(+)

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			<pre>create mode 100644 f5.txt XXX@DESKTOP-9CORRHG MINGW64 ~/lab6_advanced (main) \$ git log --oneline 31ae8ad (HEAD -> main) Step 5 9ba0339 Step 4 1389a3c Step 3 20f84cc Step 2 f742d99 Step 1</pre>
3	Создать ветку feature-rebase, добавить три коммита (Feature A, Feature B, Fix typo)	☒	<pre>XXX@DESKTOP-9CORRHG MINGW64 ~/lab6_advanced (main) \$ git checkout -b feature-rebase Switched to a new branch 'feature-rebase' XXX@DESKTOP-9CORRHG MINGW64 ~/lab6_advanced (feature-rebase) \$ echo "Feature A" > fa.txt; git add .; git commit -m "Feature A" warning: in the working copy of 'f4.txt', LF will be replaced by CRLF the next time Git touches it warning: in the working copy of 'f5.txt', LF will be replaced by CRLF the next time Git touches it warning: in the working copy of 'fa.txt', LF will be replaced by CRLF the next time Git touches it [feature-rebase a4e6bc0] Feature A 1 file changed, 1 insertion(+) create mode 100644 fa.txt XXX@DESKTOP-9CORRHG MINGW64 ~/lab6_advanced (feature-rebase) \$ echo "Feature B" > fb.txt; git add .; git commit -m "Feature B" warning: in the working copy of 'fa.txt', LF will be replaced by CRLF the next time Git touches it warning: in the working copy of 'fb.txt', LF will be replaced by CRLF the next time Git touches it [feature-rebase 362bea7] Feature B 1 file changed, 1 insertion(+) create mode 100644 fb.txt XXX@DESKTOP-9CORRHG MINGW64 ~/lab6_advanced (feature-rebase) \$ echo "Fix" >> f3.txt; git commit -am "Fix typo" warning: in the working copy of 'f3.txt', LF will be replaced by CRLF the next time Git touches it warning: in the working copy of 'fb.txt', LF will be replaced by CRLF the next time Git touches it [feature-rebase 066badc] Fix typo 1 file changed, 1 insertion(+) XXX@DESKTOP-9CORRHG MINGW64 ~/lab6_advanced (feature-rebase) \$ git log --oneline 066badc (HEAD -> feature-rebase) Fix typo 362bea7 Feature B a4e6bc0 Feature A 31ae8ad (main) Step 5 9ba0339 Step 4 1389a3c Step 3 20f84cc Step 2 f742d99 Step 1</pre>

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4	Запустить git rebase -i HEAD~3 и изменить: squash для Feature B, reword для Fix typo	☒	MINGW64:/c/Users/XXX/lab6_advanced GNU nano 9.0 C:/Users/X <pre>pick 0ab1a5f # Feature A squash 8f71b8a # Feature B reword 0b60743 # Fix typo # Rebase 55be6c3..0b60743 onto 55be6c3 (3 commands) # # Commands: # p, pick <commit> = use commit # r, reword <commit> = use commit, but edit the commit message # e, edit <commit> = use commit, but stop for amending # s, squash <commit> = use commit, but meld into previous commit # f, fixup [-C -c] <commit> = like "squash" but keep only the previous # commit's log message, unless -C is used, in which case # keep only this commit's message; -c is same as -C but # opens the editor # x, exec <command> = run command (the rest of the line) using shell # b, break = stop here (continue rebase later with 'git rebase --continue') # d, drop <commit> = remove commit # l, label <label> = label current HEAD with a name # t, reset <label> = reset HEAD to a label # m, merge [-C <commit> -c <commit>] <label> [# <oneline>] # create a merge commit using the original merge commit's # message (or the oneline, if no original merge commit was # specified); use -c <commit> to reword the commit message # u, update-ref <ref> = track a placeholder for the <ref> to be updated # to this position in the new commits. The <ref> is # updated at the end of the rebase # # These lines can be re-ordered; they are executed from top to bottom. # # If you remove a line here THAT COMMIT WILL BE LOST. # # However, if you remove everything, the rebase will be aborted. #</pre>
5	Сохранить изменения в редакторе (nano: Ctrl+O, Enter, Ctrl+X)	☒	при squash MINGW64:/c/Users/XXX/lab6_advanced GNU nano 9.0 C:/ <pre>Feature A and B combined# This is a combination of 2 commits. # This is the 1st commit message: Feature A # This is the commit message #2: Feature B # Please enter the commit message for your changes. Lines starting # with '#' will be ignored, and an empty message aborts the commit. # # Date: Sat May 16 19:10:55 2026 +0500 # # interactive rebase in progress; onto 55be6c3 # Last commands done (2 commands done): # pick 0ab1a5f # Feature A # squash 8f71b8a # Feature B # Next command to do (1 remaining command): # reword 0b60743 # Fix typo # You are currently rebasing branch 'feature-rebase' on '55be6c3'. # # Changes to be committed: # new file: fa.txt # new file: fb.txt #</pre>

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6	При squash объединить сообщения коммитов, при reword написать 'Fix critical typo in f3.txt'	✓	при reword  <pre> MINGW64/c/Users/XXX/lab6_advanced GNU nano 9.0 C: Fix critical typo in f3.txt Fix typo # Please enter the commit message for your changes. Lines starting # with '#' will be ignored, and an empty message aborts the commit. # # Date: Sat May 16 19:11:05 2026 +0500 # # interactive rebase in progress; onto 55be6c3 # Last commands done (3 commands done): # squash 8f71b8a # Feature B # reword 0b60743 # Fix typo # No commands remaining. # You are currently editing a commit while rebasing branch 'feature-rebase' on '55be6c3' # # Changes to be committed: # modified: f3.txt # </pre>
7	Проверить историю (git log --oneline) — коммитов стало меньше	✓	<pre> XXX@DESKTOP-9CORRHG MINGW64 ~/lab6_advanced (feature-rebase) \$ git log --oneline e3c30fc (HEAD -> feature-rebase) Fix critical typo in f3.txt Fix typo cb45b1f Feature A and B combined# This is a combination of 2 commits. 55be6c3 (main) Step 5 9d84d9c Step 4 a5ecb9e Step 3 ff94c7d Step 2 f82fca4 Step 1 </pre>
8	Переключиться в main и скопировать исправленный коммит через cherry-pick	✓	<pre> XXX@DESKTOP-9CORRHG MINGW64 ~/lab6_advanced (feature-rebase) \$ git checkout main Switched to branch 'main' XXX@DESKTOP-9CORRHG MINGW64 ~/lab6_advanced (main) \$ git log --oneline feature-rebase e3c30fc (feature-rebase) Fix critical typo in f3.txt Fix typo cb45b1f Feature A and B combined# This is a combination of 2 commits. 55be6c3 (HEAD -> main) Step 5 9d84d9c Step 4 a5ecb9e Step 3 ff94c7d Step 2 f82fca4 Step 1 XXX@DESKTOP-9CORRHG MINGW64 ~/lab6_advanced (main) \$ git cherry-pick e3c30fc [main b69c842] Fix critical typo in f3.txt Fix typo Date: Sat May 16 19:11:05 2026 +0500 1 file changed, 1 insertion(+) </pre>
9	Создать лёгкий тег v1.0.0 и аннотированный тег v1.1.0	✓	<pre> XXX@DESKTOP-9CORRHG MINGW64 ~/lab6_advanced (main) \$ git tag v1.0.0 XXX@DESKTOP-9CORRHG MINGW64 ~/lab6_advanced (main) \$ git tag -a v1.1.0 -m "Аннотированный тег" </pre>
10	Переключиться по тегу v1.0.0 (detached HEAD), создать от него ветку hotfix-from-tag	✓	<pre> XXX@DESKTOP-9CORRHG MINGW64 ~/lab6_advanced ((v1.1.0)) \$ git checkout -b hotfix-from-tag Switched to a new branch 'hotfix-from-tag' </pre>
11	Выполнить git reset --hard HEAD~3, затем восстановить	✓	<pre> XXX@DESKTOP-9CORRHG MINGW64 ~/lab6_advanced (main) \$ git reset --hard HEAD~3 HEAD is now at a5ecb9e Step 3 </pre>

Шаг	Описание	Вып.	Команды / вывод
	потерянные коммиты через reflog		<pre> XXX@DESKTOP-9CORRHG MINGW64 ~/lab6_advanced (main) \$ git log --oneline a5ecb9e (HEAD -> main) Step 3 ff94c7d Step 2 f82fca4 Step 1 XXX@DESKTOP-9CORRHG MINGW64 ~/lab6_advanced (main) \$ git reflog a5ecb9e (HEAD -> main) HEAD@{0}: reset: moving to HEAD~3 b69c842 (tag: v1.1.0, tag: v1.0.0, hotfix-from-tag) HEAD@{1}: checkout: moving from hotfix-from-tag to main b69c842 (tag: v1.1.0, tag: v1.0.0, hotfix-from-tag) HEAD@{2}: checkout: moving from b69c842f3e14878963013c8f284710bc69ebf22 to hotfix-from-tag b69c842 (tag: v1.1.0, tag: v1.0.0, hotfix-from-tag) HEAD@{3}: checkout: moving from main to v1.0.0 b69c842 (tag: v1.1.0, tag: v1.0.0, hotfix-from-tag) HEAD@{4}: cherry-pick: Fix critical typo in f3.txt Fix typo 55be6c3 HEAD@{5}: checkout: moving from feature-rebase to main e3c30fc (feature-rebase) HEAD@{6}: rebase (finish): returning to refs/heads/feature-rebase e3c30fc (feature-rebase) HEAD@{7}: rebase (reword): Fix critical typo in f3.txt Fix typo 7b19854 HEAD@{8}: rebase (reword): Fix typo cb45b1f HEAD@{9}: rebase (squash): Feature A and B combined# This is a combination of 2 commits. 0ab1a5f HEAD@{10}: rebase (start): checkout HEAD~3 0b60743 HEAD@{11}: commit: Fix typo 8f71b8a HEAD@{12}: commit: Feature B 0ab1a5f HEAD@{13}: commit: Feature A 55be6c3 HEAD@{14}: checkout: moving from main to feature-rebase 55be6c3 HEAD@{15}: commit: Step 5 ...skipping... a5ecb9e (HEAD -> main) HEAD@{0}: reset: moving to HEAD~3 b69c842 (tag: v1.1.0, tag: v1.0.0, hotfix-from-tag) HEAD@{1}: checkout: moving from hotfix-from-tag to main b69c842 (tag: v1.1.0, tag: v1.0.0, hotfix-from-tag) HEAD@{2}: checkout: moving from b69c842f3e14878963013c8f284710bc69ebf22 to hotfix-from-tag b69c842 (tag: v1.1.0, tag: v1.0.0, hotfix-from-tag) HEAD@{3}: checkout: moving from main to v1.0.0 b69c842 (tag: v1.1.0, tag: v1.0.0, hotfix-from-tag) HEAD@{4}: cherry-pick: Fix critical typo in f3.txt Fix typo 55be6c3 HEAD@{5}: checkout: moving from feature-rebase to main e3c30fc (feature-rebase) HEAD@{6}: rebase (finish): returning to refs/heads/feature-rebase e3c30fc (feature-rebase) HEAD@{7}: rebase (reword): Fix critical typo in f3.txt Fix typo 7b19854 HEAD@{8}: rebase (reword): Fix typo cb45b1f HEAD@{9}: rebase (squash): Feature A and B combined# This is a combination of 2 commits. 0ab1a5f HEAD@{10}: rebase (start): checkout HEAD~3 0b60743 HEAD@{11}: commit: Fix typo 8f71b8a HEAD@{12}: commit: Feature B 0ab1a5f HEAD@{13}: commit: Feature A 55be6c3 HEAD@{14}: checkout: moving from main to feature-rebase 55be6c3 HEAD@{15}: commit: Step 5 9d84d9c HEAD@{16}: commit: Step 4 </pre>

2. Самостоятельное задание

Условие:

Создайте ветку messy, сделайте 4 коммита. С помощью git rebase -i HEAD~4 объедините первые два коммита и переименуйте третий. Перенесите один коммит из messy в main через cherry-pick. Создайте аннотированный тег exercise-v1.

Выполнение (ход действий и результат):

3. Ответы на контрольные вопросы

1. Вопрос 1:

потому что rebase переписывает историю коммитов. если кто-то работает с этой веткой, то у него останется старая история, что может привести к конфликтам или двойным коммитам

2. Вопрос 2:

git cherry-pick берет конкретный коммит из любой ветки и применяет его к текущей, а git merge берет и объединяет все коммиты в текущей ветке

3. Вопрос 3:

через git reflog нужно найти хэш коммита и выполнить git reset --hard "хэш"

4. Дополнительные замечания (при необходимости)

4 вопрос:

git config --global core.editor "название редактора"

Инструкция: Выберите номер лабораторной работы. Заполняйте поля — они автоматически растягиваются по высоте под текст. Для печати нажмите кнопку «Печать» — все рамки полей исчезнут, и отчёт будет выглядеть как обычный документ.